Group 4
Allies Connect System Overview

Milestone 1

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General Architectural Foundation

- The general architecture of the site is built on a REST API for all CRUD (Create, Read, Update, Delete) operations, supported by a relational database schema (PostgreSQL or MySQL). To ensure accessibility and low overhead, the platform is mobile-first and web-based, rather than a native mobile app. The infrastructure is hosted on KSU Linux environments, utilizing a Node.js/Express backend and a React frontend to ensure a responsive experience across all devices.

General Architecture - User Access Levels

- Public Users
- Service Providers
- Admins

General Architecture - Public Resource Finder & Events

- Public Resource Finder
- Events Directory

General Architecture - Volunteer Management System

- One of the most complex structural elements is the Volunteer Management System. It is designed as a "Posting Engine" where providers can create specific opportunities linked to their organization or particular events. A major innovation here is the Hybrid Profile Data Strategy. While volunteers have a static core profile, providers can use a custom form builder to add specific questions for each opportunity, storing these unique responses in JSONB format within the database to maintain flexibility without overcomplicating the table structure.

General Architecture - Organization and Admin

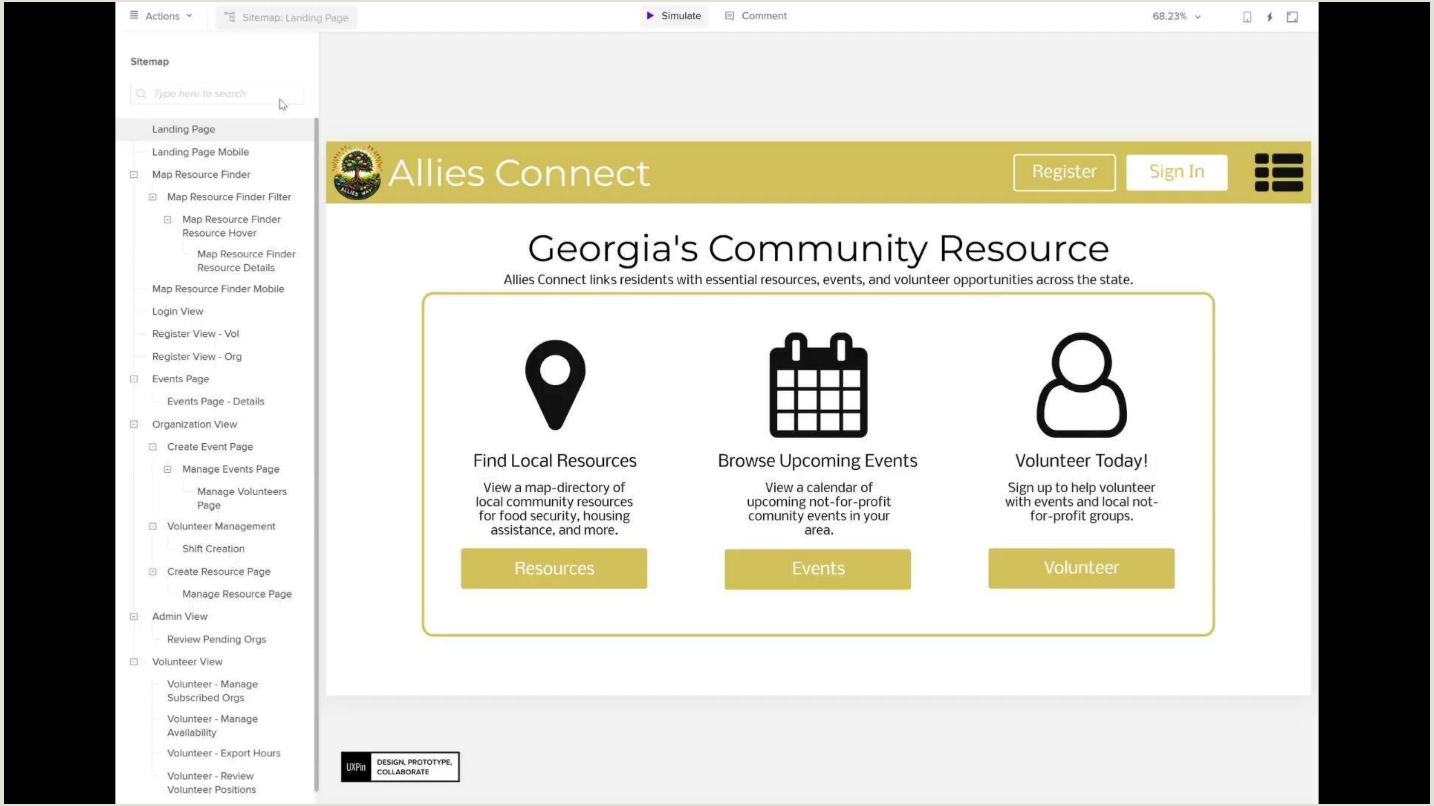
- The Organizational Portal

- The Admin Console

General Architecture -Modularity for Future Expansion

- Finally, the site structure is strictly modular. While Phase I is manual-first, the architecture is ready for Phase II automation, such as Google Calendar ICS ingestion for events and automated availability calls to verify if food pantries are currently open. This ensures that as the platform grows, the core structure remains stable while supporting new, intelligent features.

UI/UX Overview



Prioritization

Prioritization

- Not everything is likely to make it into the “Minimum Viable Product” (MVP), or “v1.0 site”
- Prioritization ensures that the most important things are included
- We’ve divided features into three categories:
  - Must have (core feature)
  - Strongly desired (the app will feel half-baked without)
  - Nice-to-have (possible to push to a later version)

Prioritization

- "Must have"
    - **Public Resource Finder:** This is the main point of the site.
    - **Back-end database and API:** We have to be able to store data.
    - **Events Directory:** Organizations frequently provide the majority of their community support through events.
    - **Access control:** Enabling the app to allow some users (such as non-logged-in people) to do some functions, and other users (such as admins) to do others is central to the app's purpose.

Prioritization

- "Strongly desired"
  - **Provider registration, EIN validation, and resource portal:** This enables providers to self-register, get validated, and manage their own data. If this isn't implemented, we can bypass this by enabling admins to handle these tasks manually.
  - **Volunteer opportunity management:** Creating volunteer opportunities, scheduling shifts, exporting volunteer data, registering as a volunteer, and signing up for event shifts are all important features of this app, but it can be launched without this feature.

Prioritization

- “Nice-to-have”
  - **Automated volunteer reminders:** Reminding people that the event they signed up to help with a month ago is important, but is distinct from the creation of the rest of the volunteer management feature.
  - **Volunteer sign-up form builder:** Being able to define the questions that are asked is a “next-level” feature that can be added on to standard questions that are always asked.
  - **Research future automation:** Creating a roadmap for future automation will help set up the v2.0 version of the site to start off with plans in place.
  - **Automation implemented:** If time allows, then actually implementing some of the automation would be some v2.0 functionality in the v1.0 site.

Data Model & Entity Relationship Diagram Overview

What We Designed

- A shared data structure for Resources, Events, and Volunteer Opportunities
- Clear separation between public browsing and account-based actions
- Support for:
  - Event Registration (RSVP)
  - Volunteer sign-ups
  - Nonprofit managed content
  - Admin oversight and activity tracking

Data Model & Entity Relationship Diagram Overview

<u>How it Works (High Level)</u>

- Users create accounts to RSVP for events or volunteer
- Nonprofits own resources, events, and volunteer opportunities
- Categories & Locations make search and filtering easy
- Relationships ensure data status accurate and consistent

<u>Why This Matters</u>

- Keeps data organized and reusable
- Makes future features easier to add
- Reduces manual work for providers
- Gives Allies Connect admin visibility and control

Automation Feasibility Research Plan

Goals

- To provide a validated and prioritized list of recommended features that should be implemented into future development phases of Allies Connect
  - Identify any repetitive processes or workflows within the system that may be suitable for automation
  - Conduct web-based research to find any automation features that are currently being used in similar systems
  - Evaluate proposed features to determine the anticipated feasibility, benefits, risk, development time and effort, and cost

Scope

- Identify and evaluate automation features relating to:
  - resource listing updates
  - resource availability verification
  - volunteer reminder and scheduling
  - data ingestion relating to calendars and events
  - data classification

Automation Feasibility Research Plan (Cont.)

Methodology

- Identify, document, and analyze how automated features are used, data ingestion is performed, and automation and user control is balanced on existing platforms to maximize system performance
- Analyze how current nonprofit and community outreach programs publish events and create and update resources to identify a standardized structure to allow for easy automation implementation

Evaluation Criteria

- Feasibility – the features compatibility with the systems design and architecture
- Cost – maintenance and monitoring needs, architecture requirements, and free/paid software and hardware package options all drive the cost of the implementation
- User Benefits – the increase in quality of the system with respect to usability, accuracy, efficiency, and effectiveness for the user
- Risk – the feature may introduce dependencies, privacy concerns, or reduced quality to the system
- Implementation Effort – ensure the feature can be created, tested, and supported in a realistic and beneficial time frame

Automation Feasibility Research Plan (Cont.)

Scoring Matrix Sample

Criterion	Score (1-5)
Feasibility	5
Cost	4
User Benefit	3
Risk	2
Implementation Effort	1

Matrix Score = 5+4+3+2+1 = 15

Automation Feasibility Research Plan (Cont.)

Validation Process

- Test the technical feasibility of the features that pass the initial screening to help identify challenges
- Features will be presented to the sponsor for feedback

Feature Selection

- Automation features will be selected based on their total criteria scores, priority categories, and overall feedback and input from the project sponsors
- There will be four priority categories:
    - Must Have - score high in all criteria, offer significant benefit, has low risk and low cost
    - Should Have - score well but has a couple of drawbacks
    - Nice to Have - has some value but also includes challenges that make it better to be considered for phases beyond Phase II
    - Not Recommended - score low because of high risk, high cost, and minimal benefit

Automation Feasibility Research Plan (Cont.)

Deliverables

- Automation Feasibility Report - document how current nonprofits publish events and resources, complete a comparative analysis of existing platforms, and create a suggested feature evaluation matrix
- Future Automation Roadmap - outline prioritized automated features for Phase II of Allies Connect with implementation plans

Success Criteria

- At least 5 feasible automation features are recommended by priority
- A complete and justified feature evaluation matrix is created
- The roadmap includes:
  - Prioritized features list
  - Recommended technical approach
  - Identified risks and mitigation
  - Suggested pilot partners for Phase II testing
- Roadmap is approved by the sponsor
- Research deliverables meet timeline and quality expectations

Let's discuss any doubts, questions or concerns.

Any questions? Ask away!

Thank you